

# Profit Dilution Resistance and the Cross Section of Stock Returns

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## Abstract

This paper studies the asset pricing implications of Profit Dilution Resistance (PDR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on PDR achieves an annualized gross (net) Sharpe ratio of 0.52 (0.45), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 24 (23) bps/month with a t-statistic of 3.12 (3.01), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Share issuance (5 year), Share issuance (1 year), Change in equity to assets, Momentum and LT Reversal, Long-run reversal) is 14 bps/month with a t-statistic of 1.99.

# 1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, yet identifying which firm characteristics expose investors to such risks remains a central challenge in finance. While the consumption-based asset pricing model provides a theoretical foundation for understanding expected returns through the lens of marginal utility and intertemporal substitution, empirical work continues to uncover cross-sectional patterns that require deeper economic interpretation (Cochrane et al., 2023). We examine Profit Dilution Resistance (PDR), a measure capturing firms’ ability to maintain profitability without diluting existing shareholders, as a characteristic that signals differential exposure to consumption risk. Firms with high PDR demonstrate resilience in maintaining profit margins while avoiding equity issuance, suggesting these firms provide better consumption hedging properties during economic downturns when marginal utility is high (Bansal and Yaron, 2004).

The economic mechanism linking PDR to expected returns operates through firms’ differential exposure to long-run consumption risks and their state-dependent performance across business cycles. Firms with high PDR maintain profitability without external equity financing, indicating operational efficiency and strong cash flow generation that correlates with aggregate consumption growth (Parker and Juliard, 2005). These characteristics make high-PDR firms particularly valuable during bad economic states when investors’ marginal utility of consumption peaks, as these firms continue generating internal resources precisely when external financing becomes costly and consumption smoothing becomes difficult (Campbell and Cochrane, 1999). The consumption-based interpretation suggests that high-PDR firms act as consumption hedges, commanding lower risk premia because they pay off when the stochastic discount factor is high.

The state dependence of PDR’s predictive power reflects its connection to the

intertemporal marginal rate of substitution and disaster risk sensitivity. During economic contractions characterized by declining consumption growth, firms resistant to profit dilution maintain their value better than those requiring frequent equity issuance, which typically occurs at depressed valuations (Wachter, 2013). This resilience translates into lower covariance with the pricing kernel, as high-PDR firms' cash flows exhibit less procyclicality with consumption innovations (Lettau and Ludwigson, 2001). The ability to avoid dilutive financing during disasters or rare consumption disasters represents a valuable real option that reduces these firms' systematic risk exposure (Barro, 2006).

Furthermore, PDR captures firms' exposure to habit-based consumption risks through its relationship with financing constraints and investment flexibility. High-PDR firms maintain investment capacity without external financing, allowing them to respond to productivity shocks that drive long-run consumption growth (Bansal et al., 2016). This operational flexibility becomes particularly valuable when consumption approaches the habit level, as investors place higher valuations on firms that can sustain operations without diluting ownership during periods of heightened marginal utility (Campbell and Cochrane, 1999). The cross-sectional variation in PDR thus reflects heterogeneous loadings on consumption-based risk factors, with high-PDR firms requiring lower expected returns due to their favorable covariance properties with the stochastic discount factor.

We found that a value-weighted long/short trading strategy based on PDR achieved an annualized gross Sharpe ratio of 0.52, placing it in the top 10% of documented anomalies in the factor zoo. The strategy delivered monthly average abnormal gross returns of 24 basis points relative to the Fama-French six-factor model, with a t-statistic of 3.12, demonstrating robust performance after controlling for market, size, value, profitability, investment, and momentum factors. The economic magnitude remained substantial after accounting for transaction costs, with net returns of 23

basis points per month and a t-statistic of 3.01, indicating that PDR’s predictive power survives realistic implementation frictions.

The PDR strategy exhibited remarkable consistency across different portfolio construction methodologies and size segments. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the PDR strategy generated average returns of 23 basis points per month with a t-statistic of 2.57, demonstrating that the effect was not confined to small, illiquid stocks where arbitrage might be limited. The strategy’s alpha relative to the six Fama-French factors ranged from 20 to 23 basis points per month among large-cap stocks, with t-statistics between 2.11 and 2.50, confirming that PDR captures a pervasive pricing phenomenon across the investable universe.

Controlling for the six most closely related anomalies from the factor zoo—including growth in book equity, share issuance measures, and momentum—plus the six Fama-French factors simultaneously, the PDR strategy maintained an alpha of 14 basis points per month with a t-statistic of 1.99. This result indicated that PDR contained incremental information about the cross-section of expected returns beyond existing characteristics. The strategy’s net Sharpe ratio of 0.45 ranked in the top 2% among 212 documented anomalies after accounting for transaction costs, while a dollar invested in the net PDR strategy would have grown to \$5.25, placing it in the top 1% of cumulative performance in the factor zoo.

Our study contributes to the consumption-based asset pricing literature by identifying PDR as a characteristic that captures firms’ differential exposure to consumption risk, extending the work on long-run risks and rare disasters in explaining the cross-section of returns (Bansal and Yaron, 2004; Barro, 2006). Unlike previous studies focusing on accounting-based profitability measures or external financing activities separately, we demonstrate that the interaction between profit generation and dilution resistance provides a more complete picture of firms’ consumption risk

exposure (Fama and French, 2015; Pontiff and Woodgate, 2008). The PDR measure synthesizes information about operational efficiency and financing policy into a single characteristic that strongly predicts returns through its connection to the stochastic discount factor.

Methodologically, we advance the literature by showing that PDR’s predictive power survives the comprehensive testing protocol of recent studies questioning the proliferation of anomalies, with our results robust to transaction costs, alternative portfolio constructions, and controls for related factors (McLean and Pontiff, 2016; Hou et al., 2020). The persistence of PDR’s alpha after controlling for established factors suggests it captures a distinct dimension of systematic risk related to firms’ ability to maintain profitability during consumption downturns. Our finding that PDR improves the mean-variance efficient frontier even after accounting for trading costs in both the test asset and benchmark factors represents a stringent test of economic significance rarely applied in the literature (Novy-Marx and Velikov, 2016).

The broader implications of our findings extend to understanding how firm characteristics map to macroeconomic risks through the consumption channel. The strong performance of PDR among large, liquid stocks suggests that markets may not fully incorporate the consumption hedging properties of profit dilution resistance into prices, potentially due to limits to arbitrage or behavioral biases in processing complex accounting information (Hirshleifer and Teoh, 2003). Our results support consumption-based models that emphasize state-dependent risk premia and highlight the importance of considering firms’ financing flexibility when assessing their exposure to systematic consumption risk, providing new empirical content for theoretical models of asset pricing under long-run risks and disaster scenarios.

## 2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and capital structures. The sample spans multiple decades of annual observations, allowing us to examine the signal's performance across various market conditions and economic cycles. We construct our Profit Dilution Resistance signal using two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet, reflecting the book value of equity capital raised through common stock issuance. Gross profit (GP) measures the difference between revenues and cost of goods sold, representing the fundamental profitability of a firm's core operations before considering operating expenses, interest, and taxes. Both variables are fundamental components of financial statements that capture distinct aspects of a firm's capital structure and operating performance. We calculate the Profit Dilution Resistance signal as the year-over-year change in common stock scaled by lagged gross profit:  $(CSTK(t) - CSTK(t-1)) / GP(t-1)$ . This ratio captures the extent to which firms issue new equity relative to their profit-generating capacity, with higher values indicating greater equity issuance relative to profitability. The economic intuition is that firms issuing substantial equity relative to their gross profit may experience earnings dilution, potentially signaling either growth opportunities requiring external financing or financial distress necessitating capital infusions. We compute the signal using fiscal year-end values and require non-missing observations for both common stock in consecutive years and lagged gross profit. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

### 3 Signal diagnostics

Figure 1 plots descriptive statistics for the PDR signal. Panel A plots the time-series of the mean, median, and interquartile range for PDR. On average, the cross-sectional mean (median) PDR is -0.03 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input PDR data. The signal's interquartile range spans -0.02 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the PDR signal for the CRSP universe. On average, the PDR signal is available for 6.28% of CRSP names, which on average make up 7.67% of total market capitalization.

### 4 Does PDR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on PDR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high PDR portfolio and sells the low PDR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short PDR strategy earns an average return of 0.31% per month with a t-statistic of 4.05. The annualized Sharpe ratio of the strategy is 0.52. The alphas range from 0.24% to 0.32% per month and have t-statistics exceeding 3.12 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.28,

with a t-statistic of 5.29 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 572 stocks and an average market capitalization of at least \$1,523 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 26 bps/month with a t-statistics of 3.45. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 22-29bps/month. The lowest return, (22 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.97. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the PDR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the PDR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and PDR, as well as average returns and alphas for long/short trading PDR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the PDR strategy achieves an average return of 23 bps/month with a t-statistic of 2.57. Among these large cap stocks, the alphas for the PDR strategy relative to the five most common factor models range from 20 to 23 bps/month with t-statistics between 2.11 and 2.50.

## 5 How does PDR perform relative to the zoo?

Figure 2 puts the performance of PDR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the PDR strategy falls in the distribution. The PDR strategy’s gross (net) Sharpe ratio of 0.52 (0.45) is greater than 90% (98%) of anomaly Sharpe ratios,

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<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the PDR strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the PDR strategy would have yielded \$7.23 which ranks the PDR strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the PDR strategy would have yielded \$5.25 which ranks the PDR strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the PDR relative to those. Panel A shows that the PDR strategy gross alphas fall between the 61 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The PDR strategy has a positive net generalized alpha for five out of the five factor models. In these cases PDR ranks between the 80 and 88 percentiles in terms of how much it could have expanded the achievable investment frontier.

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<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

## 6 Does PDR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of PDR with 210 filtered anomaly signals.<sup>3</sup> Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price PDR or at least to weaken the power PDR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of PDR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{PDR}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{PDR}PDR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{PDR,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on PDR. Stocks are finally grouped into five PDR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

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<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

PDR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on PDR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the PDR signal in these Fama-MacBeth regressions exceed 0.82, with the minimum t-statistic occurring when controlling for Momentum and LT Reversal. Controlling for all six closely related anomalies, the t-statistic on PDR is 0.00.

Similarly, Table 5 reports results from spanning tests that regress returns to the PDR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the PDR strategy earns alphas that range from 16-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.16, which is achieved when controlling for Momentum and LT Reversal. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the PDR trading strategy achieves an alpha of 14bps/month with a t-statistic of 1.99.

## 7 Does PDR add relative to the whole zoo?

Finally, we can ask how much adding PDR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the PDR signal.<sup>4</sup> We consider

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<sup>4</sup>We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes PDR grows to \$3089.32.

## 8 Conclusion

This study provides robust evidence that Profit Dilution Resistance (PDR) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that PDR-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on PDR delivers an annualized Sharpe ratio of 0.52 gross (0.45 net of transaction costs), indicating strong economic significance even after considering implementation frictions. The strategy’s ability to generate monthly abnormal returns of 24 basis points gross relative to the Fama-French five-factor model plus momentum, with a highly significant t-statistic of 3.12, underscores its robustness as a pricing anomaly. Particularly noteworthy is that PDR maintains statistical significance even when controlling for twelve factors, including six closely related strategies from the factor zoo such as share issuance metrics and reversal effects, yielding a monthly alpha of 14 basis points with a t-statistic of 1.99.

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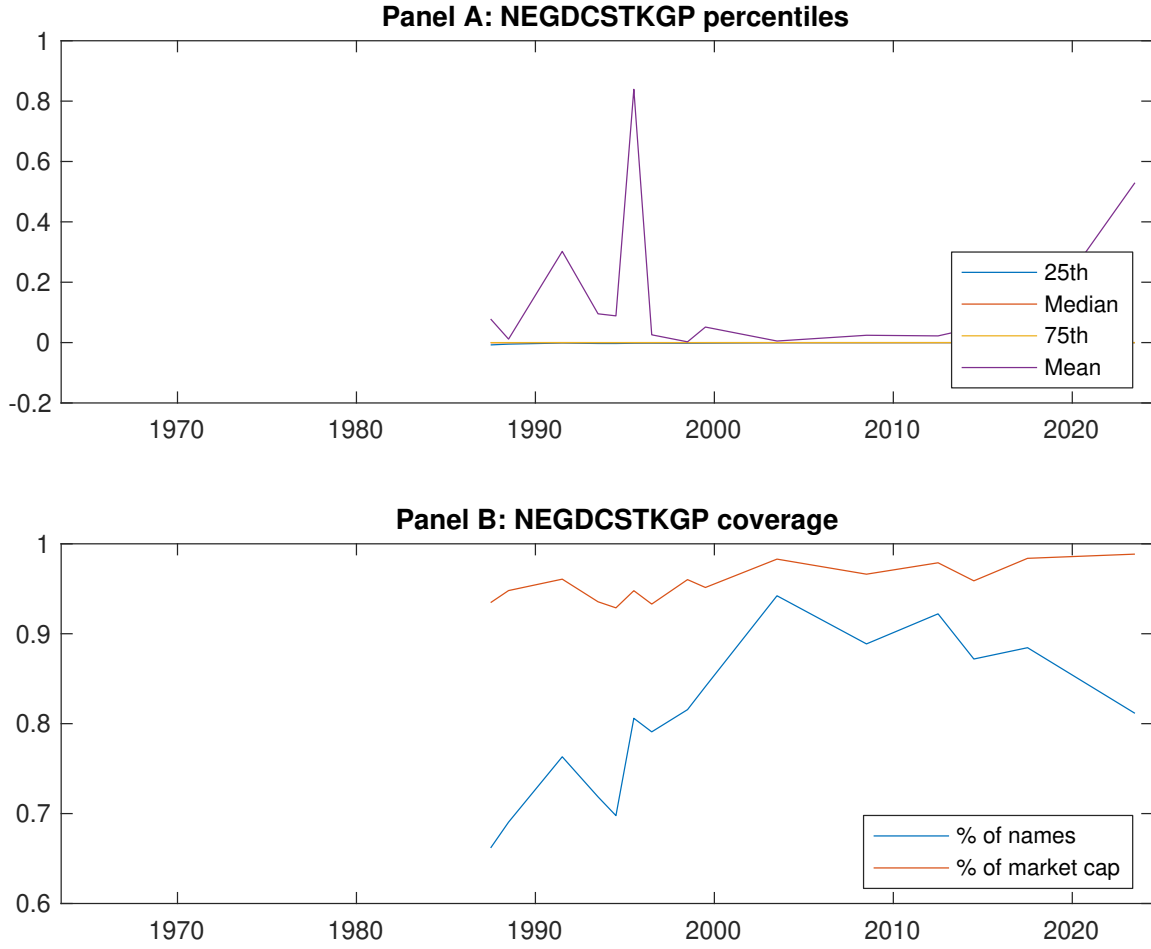
ization on CRSP in the period for which PDR is available.

These results have important practical implications for both academic research and investment practice. For academics, PDR adds to our understanding of how market participants price firms' resistance to profit dilution, suggesting that investors may systematically underreact to or misinterpret signals about firms' ability to maintain profitability in the face of growth or capital structure changes. For practitioners, the strategy's robust performance net of transaction costs and its persistence after controlling for known factors suggest it may offer genuine alpha generation opportunities, particularly when combined with other established strategies in a multi-factor framework.

Several limitations of this study warrant consideration. First, while we employ conservative transaction cost estimates, actual implementation costs may vary depending on market conditions and execution capabilities. Second, the sample period and market coverage may influence the generalizability of results, and the strategy's performance should be examined across different market regimes and international markets. Third, the economic mechanisms driving the PDR anomaly require further theoretical and empirical investigation to fully understand why this pricing inefficiency persists.

Future research should explore several promising avenues. First, investigating the behavioral or risk-based explanations for the PDR anomaly would enhance our theoretical understanding of this phenomenon. Second, examining the interaction between PDR and corporate events such as mergers, acquisitions, or significant capital raising activities could provide insights into when the signal is most informative. Third, testing the strategy's performance in international markets would establish its global relevance. Finally, exploring whether machine learning techniques can enhance the PDR signal or identify optimal combinations with other factors represents a natural extension of this work. Understanding the time-varying nature of the PDR premium and its relationship with macroeconomic conditions would also contribute

valuable insights for both risk management and tactical asset allocation decisions.



**Figure 1:** Times series of PDR percentiles and coverage.  
This figure plots descriptive statistics for PDR. Panel A shows cross-sectional percentiles of PDR over the sample. Panel B plots the monthly coverage of PDR relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on PDR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Excess returns and alphas on PDR-sorted portfolios                       |                  |                  |                  |                  |                 |                  |
|-----------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|-----------------|------------------|
|                                                                                   | (L)              | (2)              | (3)              | (4)              | (H)             | (H-L)            |
| $r^e$                                                                             | 0.44<br>[2.63]   | 0.55<br>[3.00]   | 0.61<br>[3.43]   | 0.64<br>[3.99]   | 0.74<br>[4.62]  | 0.31<br>[4.05]   |
| $\alpha_{CAPM}$                                                                   | -0.12<br>[-2.31] | -0.08<br>[-1.86] | -0.00<br>[-0.07] | 0.10<br>[1.97]   | 0.19<br>[4.35]  | 0.32<br>[4.13]   |
| $\alpha_{FF3}$                                                                    | -0.15<br>[-2.76] | -0.06<br>[-1.44] | 0.02<br>[0.37]   | 0.06<br>[1.23]   | 0.16<br>[3.61]  | 0.30<br>[3.92]   |
| $\alpha_{FF4}$                                                                    | -0.12<br>[-2.26] | -0.04<br>[-0.81] | 0.04<br>[0.89]   | 0.03<br>[0.58]   | 0.14<br>[3.30]  | 0.27<br>[3.41]   |
| $\alpha_{FF5}$                                                                    | -0.17<br>[-3.26] | 0.00<br>[0.11]   | 0.02<br>[0.49]   | -0.02<br>[-0.54] | 0.09<br>[2.20]  | 0.27<br>[3.47]   |
| $\alpha_{FF6}$                                                                    | -0.15<br>[-2.83] | 0.02<br>[0.50]   | 0.04<br>[0.92]   | -0.04<br>[-0.93] | 0.09<br>[2.11]  | 0.24<br>[3.12]   |
| Panel B: Fama and French (2018) 6-factor model loadings for PDR-sorted portfolios |                  |                  |                  |                  |                 |                  |
| $\beta_{MKT}$                                                                     | 0.97<br>[75.11]  | 1.04<br>[96.77]  | 1.02<br>[91.08]  | 1.00<br>[91.09]  | 0.98<br>[93.93] | 0.01<br>[0.59]   |
| $\beta_{SMB}$                                                                     | -0.03<br>[-1.82] | 0.00<br>[0.22]   | 0.03<br>[1.69]   | -0.06<br>[-3.98] | 0.01<br>[0.61]  | 0.04<br>[1.59]   |
| $\beta_{HML}$                                                                     | 0.10<br>[3.88]   | -0.02<br>[-0.78] | -0.06<br>[-2.96] | 0.06<br>[3.04]   | 0.01<br>[0.52]  | -0.09<br>[-2.38] |
| $\beta_{RMW}$                                                                     | 0.13<br>[5.10]   | -0.13<br>[-5.96] | 0.01<br>[0.27]   | 0.13<br>[5.89]   | 0.06<br>[3.15]  | -0.06<br>[-1.75] |
| $\beta_{CMA}$                                                                     | -0.07<br>[-1.98] | -0.12<br>[-3.86] | -0.02<br>[-0.63] | 0.17<br>[5.55]   | 0.21<br>[7.06]  | 0.28<br>[5.29]   |
| $\beta_{UMD}$                                                                     | -0.03<br>[-2.39] | -0.02<br>[-2.32] | -0.03<br>[-2.61] | 0.03<br>[2.34]   | 0.00<br>[0.33]  | 0.03<br>[1.83]   |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )       |                  |                  |                  |                  |                 |                  |
| $n$                                                                               | 684              | 652              | 572              | 671              | 803             |                  |
| $me$ (\$10 <sup>6</sup> )                                                         | 1688             | 1523             | 2173             | 2213             | 2520            |                  |

**Table 2:** Robustness to sorting methodology & trading costs

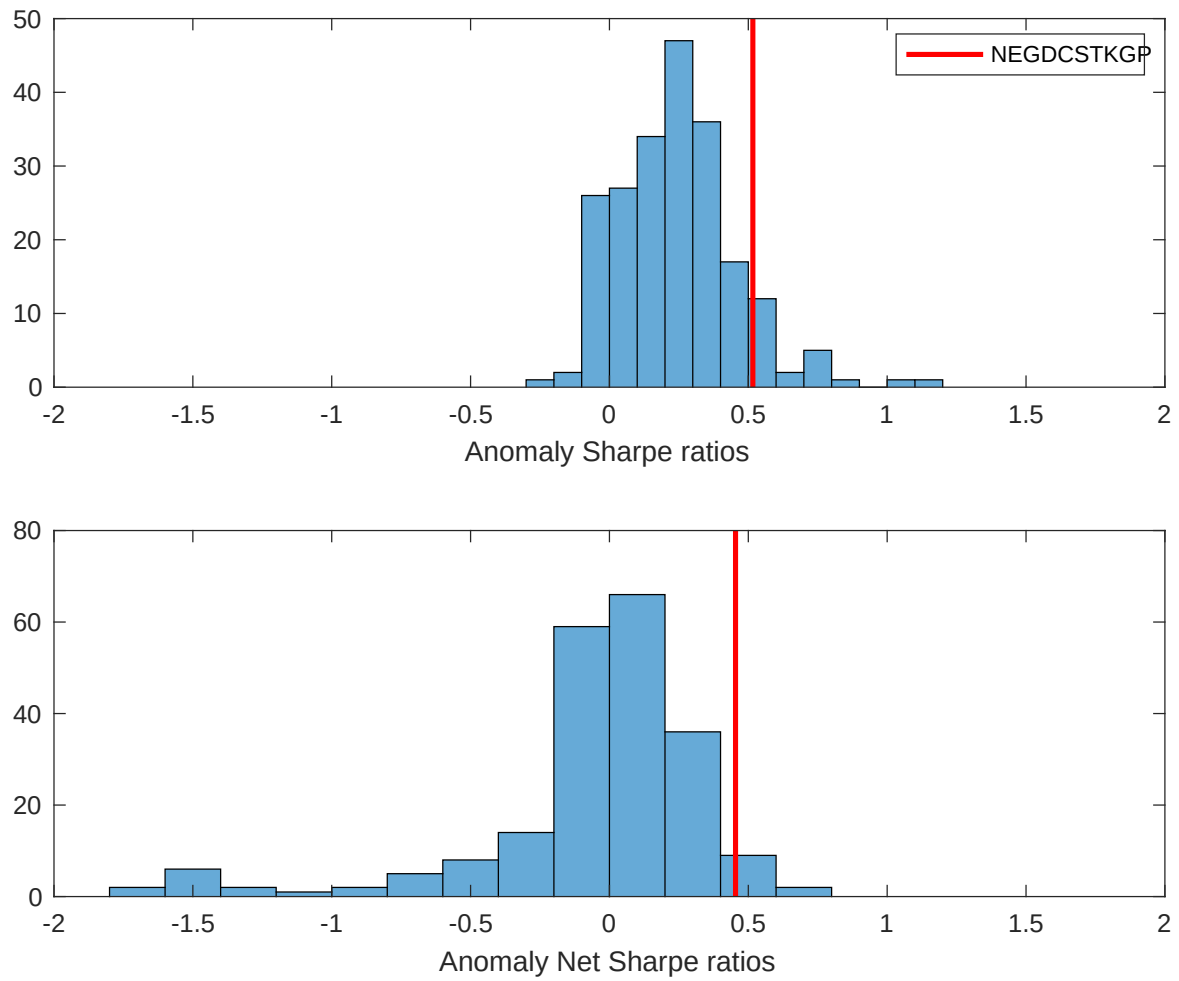
This table evaluates the robustness of the choices made in the PDR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Gross Returns and Alphas                                        |        |         |                    |                          |                         |                         |                         |                         |
|--------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                               | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                 | NYSE   | VW      | 0.31<br>[4.05]     | 0.32<br>[4.13]           | 0.30<br>[3.92]          | 0.27<br>[3.41]          | 0.27<br>[3.47]          | 0.24<br>[3.12]          |
| Quintile                                                                 | NYSE   | EW      | 0.44<br>[7.75]     | 0.46<br>[7.96]           | 0.43<br>[7.80]          | 0.38<br>[6.80]          | 0.41<br>[7.48]          | 0.37<br>[6.72]          |
| Quintile                                                                 | Name   | VW      | 0.33<br>[4.18]     | 0.34<br>[4.31]           | 0.33<br>[4.14]          | 0.30<br>[3.68]          | 0.30<br>[3.73]          | 0.28<br>[3.42]          |
| Quintile                                                                 | Cap    | VW      | 0.26<br>[3.45]     | 0.26<br>[3.48]           | 0.26<br>[3.40]          | 0.21<br>[2.78]          | 0.24<br>[3.24]          | 0.21<br>[2.78]          |
| Decile                                                                   | NYSE   | VW      | 0.33<br>[3.75]     | 0.30<br>[3.46]           | 0.29<br>[3.26]          | 0.24<br>[2.73]          | 0.28<br>[3.15]          | 0.24<br>[2.76]          |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                               | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                 | NYSE   | VW      | 0.27<br>[3.56]     | 0.28<br>[3.67]           | 0.26<br>[3.46]          | 0.24<br>[3.18]          | 0.24<br>[3.20]          | 0.23<br>[3.01]          |
| Quintile                                                                 | NYSE   | EW      | 0.25<br>[3.94]     | 0.28<br>[4.36]           | 0.25<br>[3.99]          | 0.22<br>[3.59]          | 0.21<br>[3.55]          | 0.20<br>[3.27]          |
| Quintile                                                                 | Name   | VW      | 0.29<br>[3.70]     | 0.30<br>[3.90]           | 0.29<br>[3.71]          | 0.27<br>[3.46]          | 0.27<br>[3.47]          | 0.26<br>[3.31]          |
| Quintile                                                                 | Cap    | VW      | 0.22<br>[2.97]     | 0.22<br>[3.03]           | 0.22<br>[2.93]          | 0.19<br>[2.60]          | 0.21<br>[2.94]          | 0.20<br>[2.69]          |
| Decile                                                                   | NYSE   | VW      | 0.28<br>[3.26]     | 0.26<br>[2.94]           | 0.24<br>[2.76]          | 0.22<br>[2.49]          | 0.23<br>[2.72]          | 0.22<br>[2.52]          |

**Table 3:** Conditional sort on size and PDR

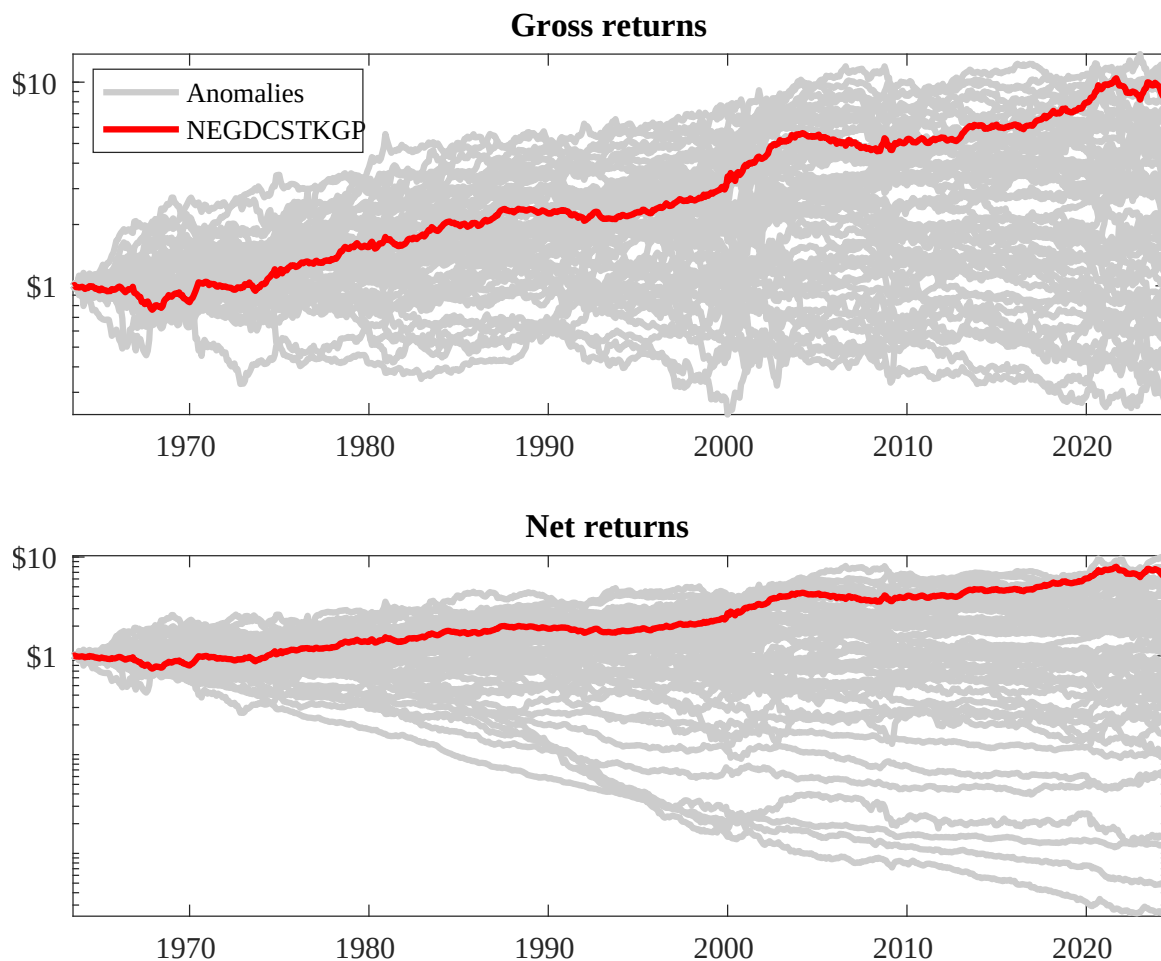
This table presents results for conditional double sorts on size and PDR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on PDR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high PDR and short stocks with low PDR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

| Panel A: portfolio average returns and time-series regression results |               |                |                |                |                |                                                    |                |                 |                |                |                |                |
|-----------------------------------------------------------------------|---------------|----------------|----------------|----------------|----------------|----------------------------------------------------|----------------|-----------------|----------------|----------------|----------------|----------------|
| Size quintiles                                                        | PDR Quintiles |                |                |                |                | PDR Strategies                                     |                |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)                                                | $r^e$          | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |
|                                                                       | (1)           | 0.47<br>[1.95] | 0.77<br>[3.10] | 0.98<br>[3.86] | 0.92<br>[3.85] | 0.95<br>[3.73]                                     | 0.50<br>[5.01] | 0.50<br>[4.96]  | 0.54<br>[5.48] | 0.42<br>[4.34] | 0.61<br>[6.32] | 0.51<br>[5.37] |
|                                                                       | (2)           | 0.63<br>[2.84] | 0.77<br>[3.31] | 0.89<br>[3.80] | 0.94<br>[4.24] | 0.84<br>[3.79]                                     | 0.21<br>[2.48] | 0.23<br>[2.66]  | 0.21<br>[2.53] | 0.17<br>[1.98] | 0.26<br>[3.06] | 0.22<br>[2.60] |
|                                                                       | (3)           | 0.55<br>[2.79] | 0.67<br>[3.12] | 0.81<br>[3.74] | 0.81<br>[4.01] | 0.90<br>[4.53]                                     | 0.35<br>[4.59] | 0.34<br>[4.45]  | 0.34<br>[4.39] | 0.30<br>[3.76] | 0.35<br>[4.48] | 0.32<br>[3.96] |
|                                                                       | (4)           | 0.47<br>[2.52] | 0.67<br>[3.34] | 0.81<br>[4.06] | 0.75<br>[4.00] | 0.80<br>[4.37]                                     | 0.33<br>[4.51] | 0.34<br>[4.64]  | 0.31<br>[4.20] | 0.30<br>[4.03] | 0.19<br>[2.68] | 0.20<br>[2.75] |
|                                                                       | (5)           | 0.47<br>[2.89] | 0.49<br>[2.70] | 0.55<br>[3.15] | 0.50<br>[3.07] | 0.70<br>[4.42]                                     | 0.23<br>[2.57] | 0.23<br>[2.50]  | 0.23<br>[2.47] | 0.20<br>[2.11] | 0.22<br>[2.42] | 0.20<br>[2.16] |
| Panel B: Portfolio average number of firms and market capitalization  |               |                |                |                |                |                                                    |                |                 |                |                |                |                |
| Size quintiles                                                        | PDR Quintiles |                |                |                |                | PDR Quintiles                                      |                |                 |                |                |                |                |
|                                                                       | Average $n$   |                |                |                |                | Average market capitalization (\$10 <sup>6</sup> ) |                |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)                                                | (L)            | (2)             | (3)            | (4)            | (H)            |                |
|                                                                       | (1)           | 375            | 376            | 374            | 373            | 366                                                | 31             | 33              | 38             | 28             | 29             |                |
|                                                                       | (2)           | 108            | 106            | 107            | 107            | 105                                                | 57             | 55              | 57             | 56             | 54             |                |
|                                                                       | (3)           | 78             | 77             | 77             | 77             | 77                                                 | 98             | 95              | 98             | 99             | 98             |                |
|                                                                       | (4)           | 65             | 65             | 65             | 65             | 65                                                 | 205            | 206             | 216            | 210            | 214            |                |
| (5)                                                                   | 60            | 60             | 60             | 60             | 60             | 1447                                               | 1524           | 1677            | 1622           | 1876           |                |                |



**Figure 2:** Distribution of Sharpe ratios.

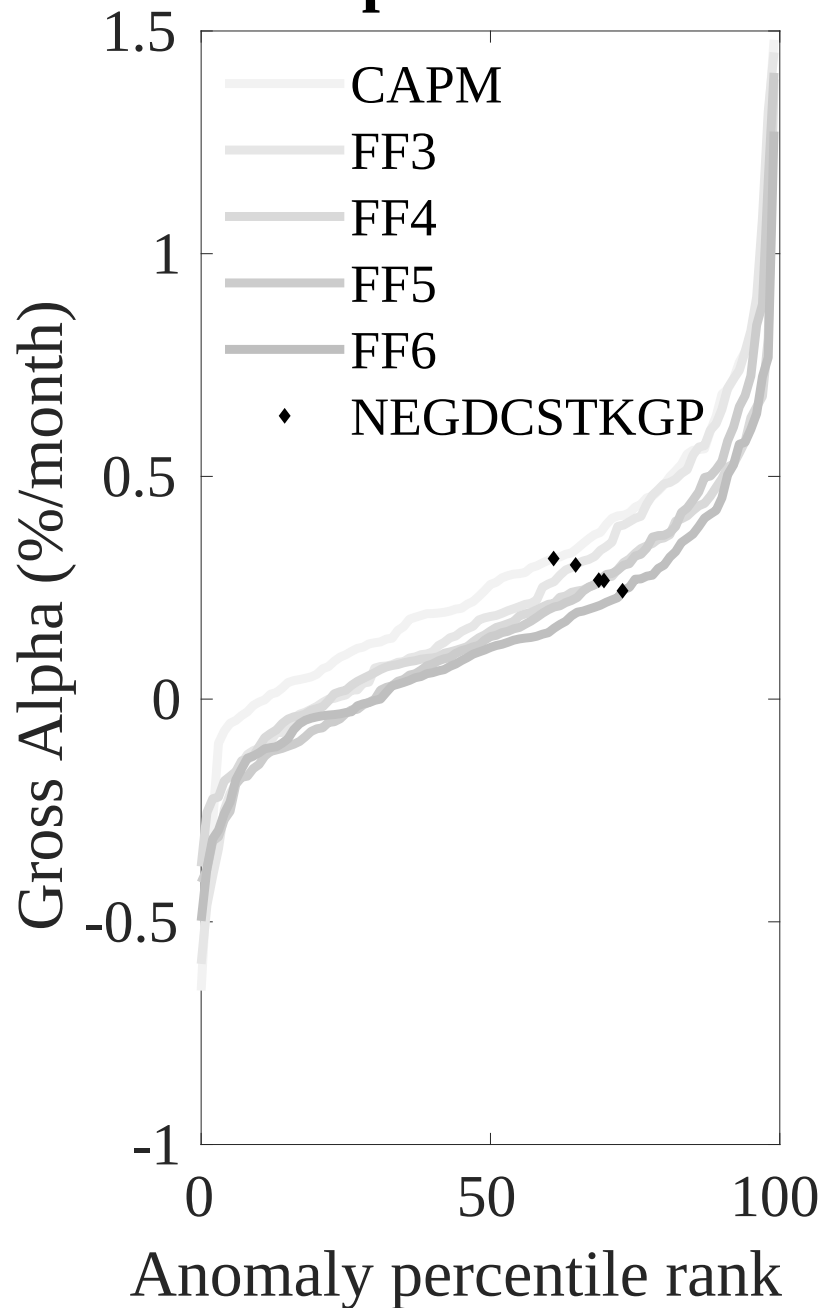
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the PDR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



**Figure 3:** Dollar invested.

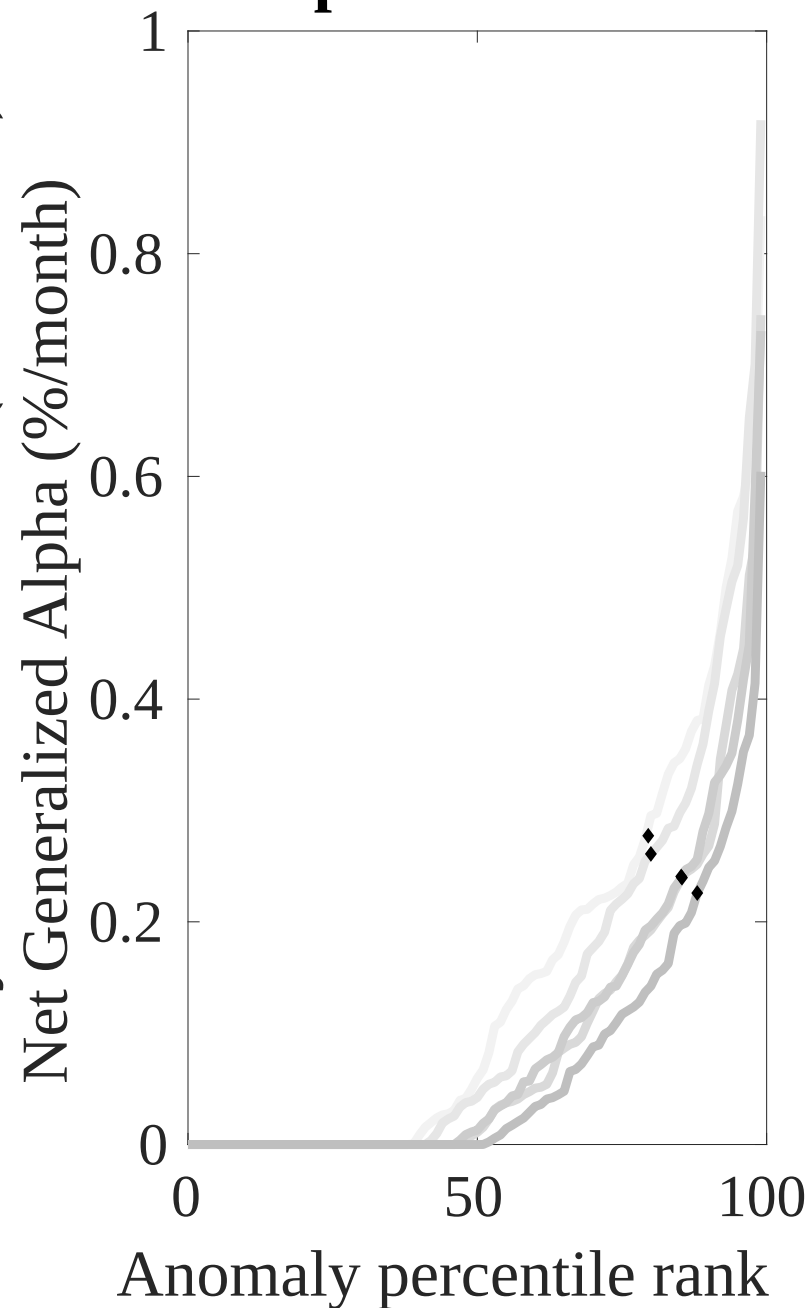
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the PDR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

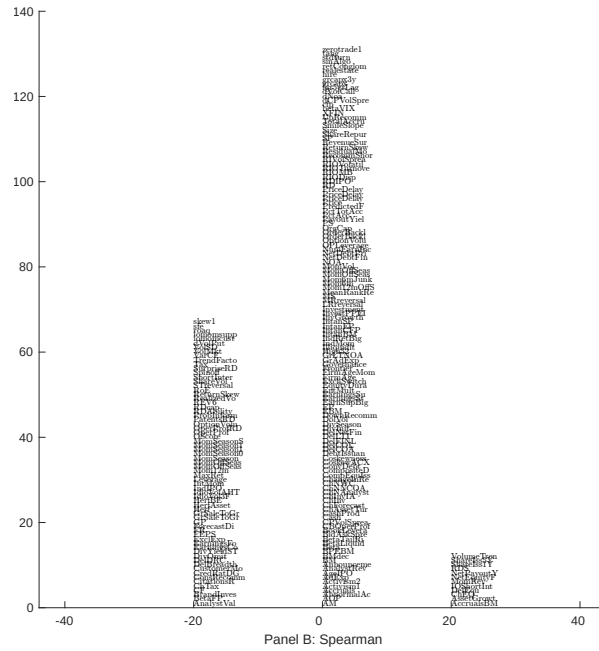
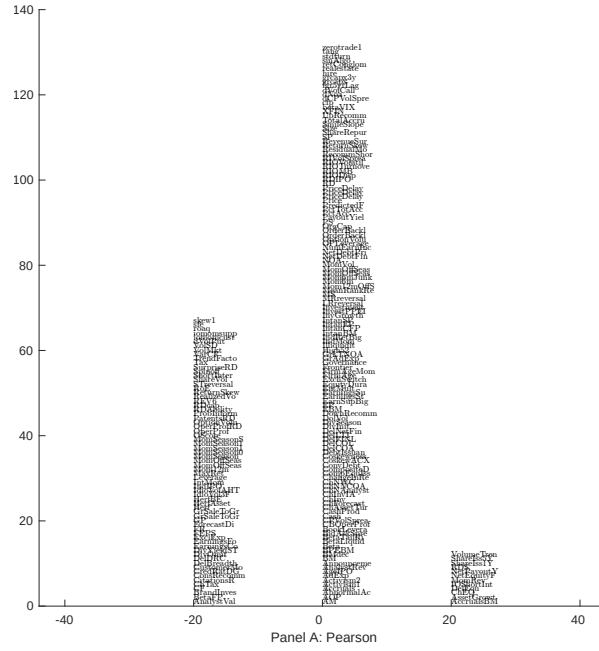
## Gross Alpha Percentiles



## Net Alpha Percentiles

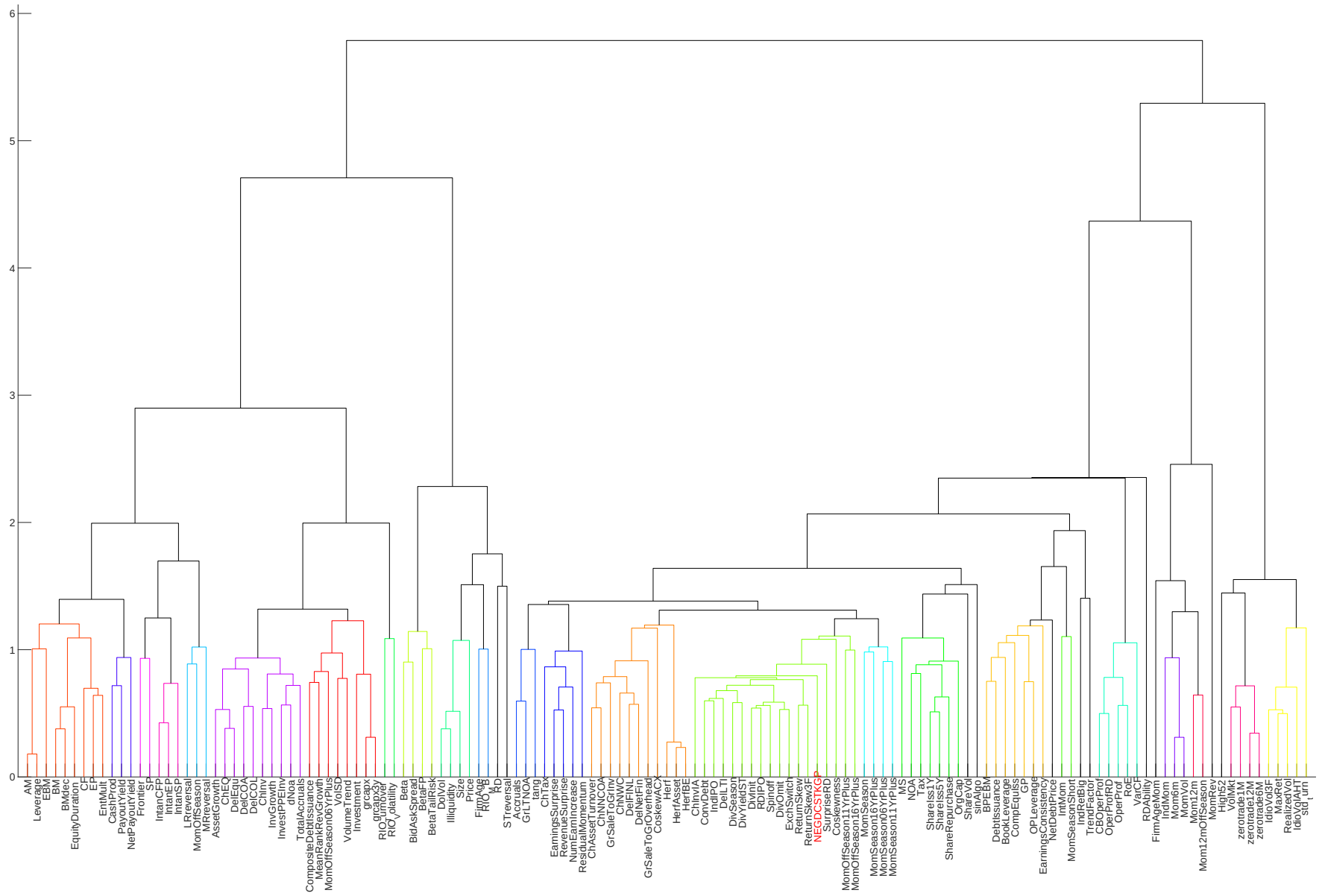
Novy-Marx and Velikov (RFS, 2016)





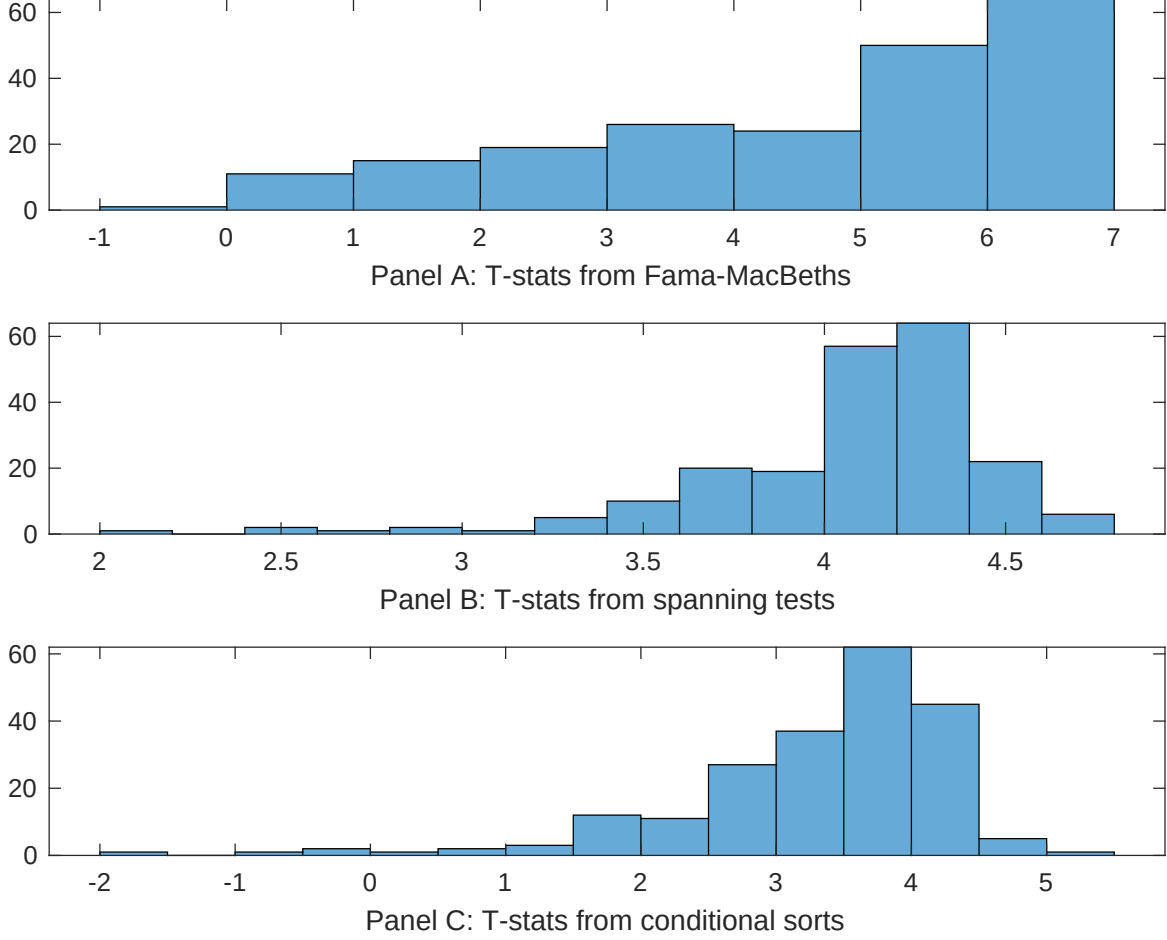
**Figure 5:** Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with PDR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.



**Figure 6:** Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.



**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of PDR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{PDR}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{PDR}PDR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{PDR,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on PDR. Stocks are finally grouped into five PDR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PDR trading strategies conditioned on each of the 210 filtered anomalies.

**Table 4:** Fama-MacBeths controlling for most closely related anomalies

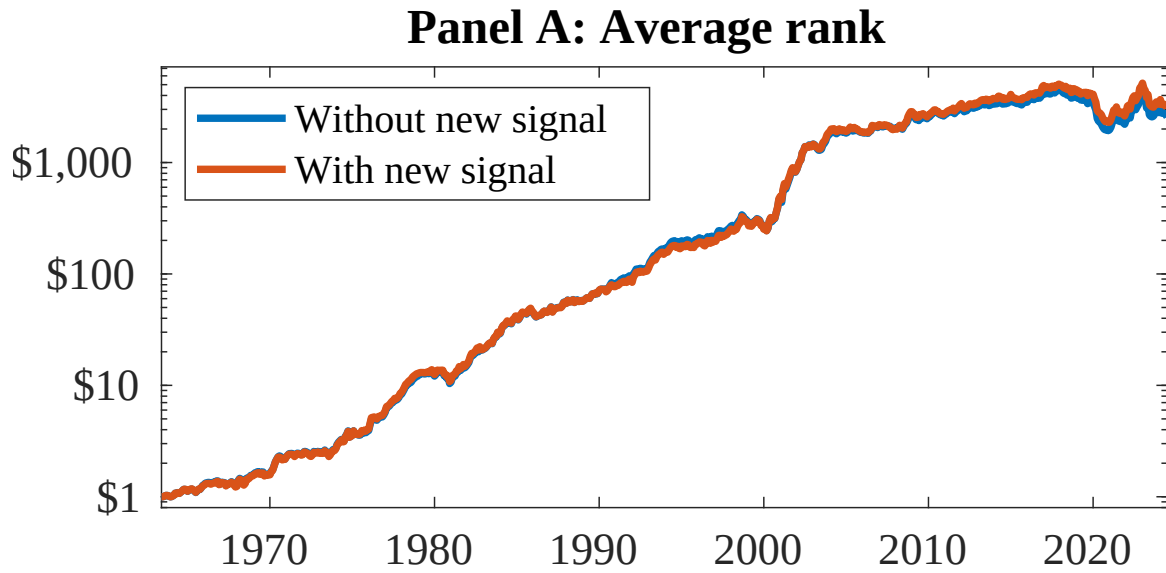
This table presents Fama-MacBeth results of returns on PDR. and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{PDR}PDR_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are Growth in book equity, Share issuance (5 year), Share issuance (1 year), Change in equity to assets, Momentum and LT Reversal, Long-run reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

|                 |                |                |                |                |                |                |                  |
|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|
| Intercept       | 0.17<br>[6.93] | 0.13<br>[6.46] | 0.13<br>[6.09] | 0.12<br>[5.77] | 0.40<br>[1.19] | 0.12<br>[5.83] | 0.10<br>[2.50]   |
| PDR             | 0.14<br>[4.69] | 0.11<br>[3.80] | 0.12<br>[4.45] | 0.15<br>[5.52] | 0.10<br>[0.82] | 0.14<br>[5.57] | 0.26<br>[0.00]   |
| Anomaly 1       | 0.38<br>[3.19] |                |                |                |                |                | 0.30<br>[1.25]   |
| Anomaly 2       |                | 0.26<br>[4.76] |                |                |                |                | 0.28<br>[2.83]   |
| Anomaly 3       |                |                | 0.13<br>[5.79] |                |                |                | 0.11<br>[2.21]   |
| Anomaly 4       |                |                |                | 0.12<br>[3.23] |                |                | -0.11<br>[-1.46] |
| Anomaly 5       |                |                |                |                | 0.11<br>[4.24] |                | 0.97<br>[3.01]   |
| Anomaly 6       |                |                |                |                |                | 0.22<br>[2.49] | -0.26<br>[-0.20] |
| # months        | 720            | 715            | 715            | 720            | 648            | 715            | 600              |
| $\bar{R}^2(\%)$ | 0              | 0              | 0              | 0              | 2              | 1              | 0                |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the PDR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{PDR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are Growth in book equity, Share issuance (5 year), Share issuance (1 year), Change in equity to assets, Momentum and LT Reversal, Long-run reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

|                 |                   |                   |                   |                  |                  |                  |                   |
|-----------------|-------------------|-------------------|-------------------|------------------|------------------|------------------|-------------------|
| Intercept       | 0.23<br>[3.06]    | 0.16<br>[2.16]    | 0.23<br>[3.03]    | 0.24<br>[3.15]   | 0.22<br>[2.88]   | 0.25<br>[3.24]   | 0.14<br>[1.99]    |
| Anomaly 1       | 29.44<br>[7.27]   |                   |                   |                  |                  |                  | 29.60<br>[5.12]   |
| Anomaly 2       |                   | 32.34<br>[8.15]   |                   |                  |                  |                  | 20.89<br>[4.57]   |
| Anomaly 3       |                   |                   | 28.17<br>[7.44]   |                  |                  |                  | 11.92<br>[2.70]   |
| Anomaly 4       |                   |                   |                   | 11.17<br>[2.93]  |                  |                  | -13.71<br>[-2.65] |
| Anomaly 5       |                   |                   |                   |                  | 4.80<br>[4.61]   |                  | 3.95<br>[3.71]    |
| Anomaly 6       |                   |                   |                   |                  |                  | 5.07<br>[2.24]   | -0.34<br>[-0.14]  |
| mkt             | 2.27<br>[1.27]    | 2.73<br>[1.54]    | 1.99<br>[1.12]    | 1.10<br>[0.60]   | 1.45<br>[0.80]   | 1.04<br>[0.56]   | 3.82<br>[2.21]    |
| smb             | 4.58<br>[1.78]    | 7.70<br>[2.99]    | 6.63<br>[2.57]    | 5.42<br>[2.04]   | 3.62<br>[1.36]   | 3.28<br>[1.17]   | 5.38<br>[2.07]    |
| hml             | -10.75<br>[-3.09] | -4.68<br>[-1.37]  | -7.96<br>[-2.31]  | -8.60<br>[-2.40] | -8.36<br>[-2.37] | -8.65<br>[-2.37] | -8.93<br>[-2.63]  |
| rmw             | -5.06<br>[-1.45]  | -18.09<br>[-4.84] | -21.74<br>[-5.38] | -5.43<br>[-1.51] | -4.01<br>[-1.12] | -4.33<br>[-1.17] | -18.44<br>[-4.54] |
| cma             | -1.47<br>[-0.23]  | 11.43<br>[2.13]   | 9.58<br>[1.72]    | 15.80<br>[2.44]  | 24.69<br>[4.76]  | 23.10<br>[4.22]  | -6.78<br>[-1.04]  |
| umd             | 3.08<br>[1.74]    | 1.47<br>[0.83]    | 2.30<br>[1.30]    | 3.88<br>[2.13]   | -1.08<br>[-0.52] | 3.90<br>[2.14]   | -3.07<br>[-1.52]  |
| # months        | 732               | 728               | 728               | 732              | 728              | 728              | 728               |
| $\bar{R}^2(\%)$ | 12                | 13                | 12                | 7                | 8                | 6                | 20                |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as PDR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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